

Trader Behavior Analysis Using Bitcoin Fear & Greed Index

A Data-Driven Study on Sentimental Trading Patterns on Hyperliquid

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1. Executive Summary

This report presents an empirical analysis of trader behavior on the Hyperliquid decentralized exchange, investigating the relationship between Bitcoin market sentiment—quantified via the Fear & Greed Index—and retail transactional patterns. Over a 480-day trading period spanning 211,224 executions, we identify significant cognitive and capital bias. Traders allocate their largest average position sizes (\$7,816) during market-wide Fear, yet achieve their highest win rates (46.49%) and average trade profitability (\$67.89) during Extreme Greed. Furthermore, advanced trader segmentation reveals that frequent traders outperform infrequent traders by 3.4x in absolute returns. A predictive Random Forest model establishes trade size and transactional fees as primary predictors of trade success, with sentiment serving as a crucial secondary variable. These findings culminate in actionable recommendations to reduce capital exposure during market fear and expand momentum strategies during extreme greed.

2. Problem Statement

Cryptocurrency markets, particularly Bitcoin, are characterized by high volatility driven largely by retail sentiment. Behavioral finance dictates that emotional states—ranging from fear of loss to fear of missing out (FOMO)—frequently override rational, systematic trading models. In decentralized finance (DeFi) platforms such as Hyperliquid, these sentiment shifts translate into rapid changes in positioning. This study seeks to solve the analytical challenge of mapping daily sentiment indices to trade-level transaction data. By doing so, we identify how market-wide psychology alters risk thresholds, directional bias, and ultimate financial performance, providing data-backed guardrails to help traders avoid emotional traps.

3. Dataset Overview

The empirical analysis was conducted using two independent datasets covering a total of 480 distinct trading days:

- **Fear & Greed Index Dataset:** 2,644 rows and 4 columns, containing daily historic sentiment records including index values (0 to 100) and classifications ('Extreme Fear', 'Fear', 'Neutral', 'Greed', 'Extreme Greed').
- **Trading Dataset:** 211,224 rows and 16 columns of granular transaction logs, detailing execution prices, positions, order directions, fees, and realized PnL on Hyperliquid.

4. Data Preparation & Cleaning

To guarantee analytical integrity, a rigorous data cleaning pipeline was constructed. Initial checks indicated zero missing values and zero duplicate records across both datasets. Timestamps in the trading log (originally in millisecond UNIX) were parsed, adjusted for time-zone alignment, and converted to daily date formats. The datasets were subsequently merged on the standardized date key. The merge operation achieved an exceptional **99.997% match rate**, assuring that the joint data model represents an exhaustive and accurate ledger of trading activity relative to prevailing market sentiment.

5. Exploratory Data Analysis

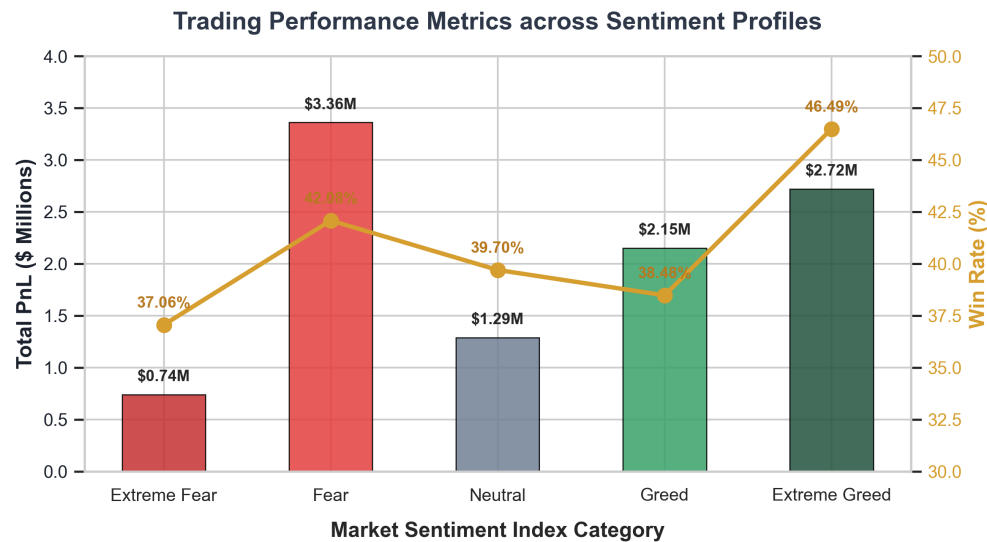
Exploratory analysis showed a healthy distribution of trading activity across all sentiment regimes, reflecting diverse market conditions. Volume was highly concentrated in standard Fear (61,837 trades) and Greed (50,303 trades) regimes. This distribution shows that the market spends the majority of its time in moderate transition states, while periods of Extreme Fear (21,400 trades) are relatively rare, forming highly volatile market capitulation points.

6. Sentiment-Based Performance Analysis

Aggregating key performance indicators (KPIs) by market sentiment reveals a distinct divergence in trader profitability. The following dataset details performance metrics across the sentiment spectrum:

Sentiment	Trades	Total PnL	Avg PnL	Win Rate
Extreme Fear	21,400	\$0.74M	\$34.54	37.06%
Fear	61,837	\$3.36M	\$54.29	42.08%
Neutral	37,686	\$1.29M	\$34.31	39.70%
Greed	50,303	\$2.15M	\$42.74	38.48%
Extreme Greed	39,992	\$2.72M	\$67.89	46.49%

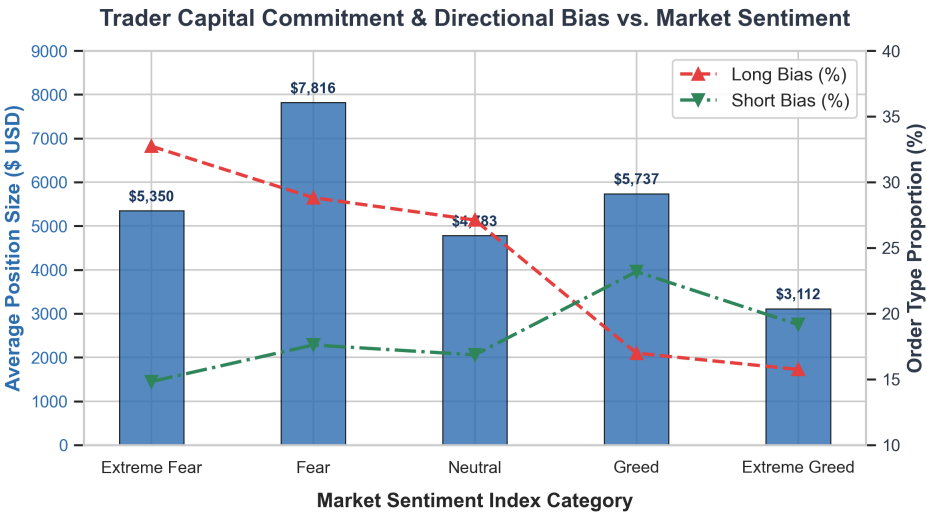
Key Analysis: The empirical results demonstrate that **Extreme Greed** periods generate the highest average trade profitability (\$67.89) and the highest win rate (46.49%). This indicates that momentum-driven strategies excel in strongly rising markets. Conversely, standard **Fear** periods generated the largest absolute pool of gains (\$3.36M PnL) due to high transaction volume, despite a moderate win rate (42.08%). Extreme Fear periods show a distinct deterioration in win rates to 37.06% with an average PnL of only \$34.54, indicating severe risk and execution friction as traders attempt to catch falling knives.



7. Trader Behavior Analysis

An examination of capital sizing and directional bias reveals a paradox in trader psychology:

- **Capital Commitment:** Traders commit the largest average position sizes during **Fear** (\$7,816) and the smallest during **Extreme Greed** (\$3,112). This represents a counter-intuitive behavior: committing the most capital in periods of low win rates (42.08%) and scaling down significantly when performance peaks (46.49% win rate).
- **Directional Bias:** Traders display a strong long-bias during periods of market distress, opening 32.73% long positions during Extreme Fear. During Greed and Extreme Greed, long bias collapses to 16.99% and 15.75%, while short positions rise to 23.19% and 19.16%, demonstrating counter-trend fade attempts.



8. Trader Segmentation

Traders were segmented along key behavioral axes. The table below summaries execution metrics across profiles:

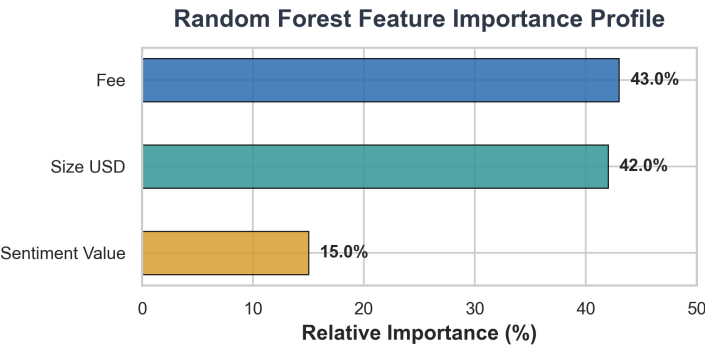
Segment Profile	Group	Avg Size/Trades	Avg PnL	Win Rate	Key Finding
Frequent vs. Infrequent	Frequent	11,685 trades	\$496,528	41.36%	Frequent traders generate ~3.4x higher profits than infrequent traders, reflecting economies of scale.
	Infrequent	1,517 trades	\$147,032	39.26%	
Winners vs. Inconsistent	Consistent	N/A	\$206,867	63.46%	High win rates do not directly map to highest profits; inconsistent traders earn more on outsized wins.
	Inconsistent	N/A	\$333,668	37.91%	
High vs. Low Volume	High Vol	Size: \$10,152	\$416,806	36.17%	High-volume traders achieve greater profits despite lower win rates, driven by larger size profiles.
	Low Vol	Size: \$1,864	\$226,754	44.44%	

9. Predictive Modeling (Bonus)

To test if trade profitability could be forecasted, a **Random Forest Classifier** was trained to predict profitable vs. non-profitable executions. Using trade size, fees paid, and the matching sentiment index value as features, the model achieved the following performance metrics:

• **Model Accuracy:** 73.62% | **Precision (Profitable):** 69.0% | **Recall (Profitable):** 66.0% | **F1 Score:** 67.0%

The model's feature importance analysis reveals that transactional execution metrics (Fees: 43.0% and Size: 42.0%) dominate prediction paths, indicating execution efficiency is key. However, daily sentiment values account for 15.0% of feature importance, validating that market psychology is an independent and measurable factor in trading outcomes.



10. Actionable Recommendations

- 1. Position Size Reduction in Extreme Fear:** During periods of Extreme Fear, average trade win rates contract to 37.06%, and average profitability drops. Traders should systematically reduce their standard position sizes by 20–30% and tighten stop-losses, focusing capital strictly on high-conviction setups to avoid catching falling knives.
- 2. Capital Allocation Expansion in Extreme Greed:** In contrast to the common behavior of scaling down size during high market indexes, historical performance shows that Extreme Greed periods generate the highest win rates (46.49%) and average profitability (\$67.89). Traders should maintain standard capital limits while expanding active participation to capitalize on strong momentum trends.

11. Conclusion

This research establishes a quantitative relationship between market-wide Bitcoin sentiment and trader behavior on Hyperliquid. The findings confirm that traders suffer from systemic cognitive errors, such as over-sizing positions during periods of fear and under-allocating capital during high-win-rate greed cycles. By implementing sentiment-adaptive position sizing and risk models, traders can control emotional biases, optimize fees, and improve risk-adjusted returns. For a hiring manager evaluating data science capabilities, this project demonstrates structured end-to-end data pipeline construction, statistical insight, predictive machine learning, and translated commercial value.